

Question Number	Answer	Additional guidance	Mark
8(a)	An explanation that includes three of the following points: □ (habitat loss) due to {deforestation / urbanisation / water drainage} (1) □ reduction in food due to {habitat loss / competition with other species / pollution} (1) □ (reduction in population) due to {hunting / fewer mates / disease} (1) □ introduction of new predator (by humans) (1)	accept pollution of habitat e.g. water pollution, pesticides accept outcompeted by other species	(3)

Question Number	Answer	Additional guidance	Mark
8(b)	An explanation that includes four of the following points: • (genetic) mutation occurred (1) • (new) allele (coding for protein) involved in fungal resistance (1) • frogs resistant to the fungus {survive and breed / have selective advantage} (1) • pass this (resistant) allele onto the offspring (1) • increase in resistant allele frequency (1)	ignore gene ignore immunity Accept clear statement that fungus is acting as selection pressure	(4)

Question Number	Answer	Additional guidance	Mark
8(c)	A description that includes the following points: • {analyse the DNA of (Sehuencas water) frogs / determine the number of heterozygotes} / determine the population size of (Bolivian Sehuencas water) frogs (1) • divide the number of heterozygotes by the total number of (Sehuencas water) frogs (in the Bolivia population) (1)		(2)

Question Number	Answer
8 (d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • genetic diversity identified as the variety of alleles in the population • genetic analysis of the frog in the zoo and the frogs in Bolivia / method of analysis • consideration of maintaining genetic {diversity / variation} / reducing inbreeding / {maintain / increase} gene pool • captive breeding / breeding in zoos or in natural habitat • use of stud books (without ref to maintaining genetic diversity) • {conservation of / increase} frog habitat in Bolivia • guidance to local government / raising awareness of need for protected area e.g. national park • conservation in zoos / recreating similar habitat in zoos / protection from predators (in zoos) / food provision • eradication / treatment of {fungal / other} disease • protection from predators (in-situ) • reintroduction of frogs from captive breeding programme • suggested details of method of reintroduction • education of local population in {Bolivia / country where zoo is located}

	research			(6)
			Additional guidance	
Level 0	0	No awardable content		
Level 1	1-2	Demonstrates isolated elements of biological knowledge related to the given context with generalised comments made. The description will contain basic information with some attempt made to link knowledge and understanding to the given context.	level 1 1 mark – one aspect described 2 marks – two aspects described	
Level 2	3-4	Demonstrates adequate knowledge by selecting and applying some relevant biological facts/concepts to provide the description being presented. The description shows some linkages and lines of reasoning with some structure.	3 marks – three aspects described 4 marks – four aspects described	
Level 3	5-6	Demonstrates comprehensive knowledge by selecting and applying relevant knowledge of biological facts/concepts to provide the description being presented. The description is clear, coherent and logically structured.	5 marks – five aspects described 6 marks – six aspects described	